Aggregate Effects of Workfare Programs

Arnav Deshpande*

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Preliminary and Incomplete

1. Introduction

Cash-for-work ('workfare') programs are pervasive in developing countries (World Bank (2015)). These programs involve providing income in return for short-term, manual labor on public works projects. Although widely used, the macroeconomic impacts of cash-for-work are not well understood. In this paper, I am interested in the aggregate effects of workfare in a setting where employers have wage-setting power. Towards this end, I study workfare in a static model of oligopsonistic competition with endogenously variable wage markdowns.

I am motivated by recent evidence from a large-scale workfare expansion in India, where the world's largest workfare program is run. It was found that in regions where the workfare program expanded, employment and wages in the market economy (the economy excluding workfare) went up. Specifically, a 30% expansion in workfare employment drove a 10% increase in the market wage and an 18% increase in market employment. It was also found that employment went up more in more concentrated labor markets. I characterize conditions under which a specific model featuring oligopsonistic competition and workfare can reproduce both these results. Thereafter, I calibrate the model to match salient features of the rural Indian economy, and evaluate the quantitative effects of a workfare expansion in the model. Simulating a workfare expansion of the same size as in India (30%) predicts 1.1% to 3.9% increase in market wages and 1.4% to 4.3% increase in market employment. The model effects lie within the empirical confidence intervals corresponding to the significance level at which the estimates are statistically significant.

How do we interpret this result? The causal evidence from the workfare expansion is consistent with a model featuring non-competitive labor markets; but it is also consistent with models featuring competitive labor markets. For one, a 30% workfare expansion can generate an increase in demand for goods, driving an increase in demand for labor, and consequently, an increase in employment and wages. It is also found that there is an increase in uptake of loans following the workfare expansion. An increase in demand for labor thus could also be driven by a relaxation of the financial constraints. I write a model that features only labor market power; and thus shuts down alternative drivers of employment and wage effects. The model isolates the quantitative importance of the market power channel in explaining the causal evidence. This excercise is informative for a planner who is choosing between unconditional cash transfers and workfare programs as an instrument for redistribution; since cash transfers would also achieve an increase in demand and a relaxation of financial constraints. Through the lens of the model developed in this paper, the market power channel is not first order in justifying workfare.

*NYU, ad6790@nyu.edu.

One might reasonably argue that it is not clear if the mechanisms embedded in this specific model of market power are the right mechanisms. Within the macroeconomics literature, about three broad origins of labor market power have been explored. The approach adopted in this paper specifies preferences over jobs. It builds on Berger et al. (2022), who use an Atkeson and Burstein (2008) style model of product market power (Edmond et al. (2015)) and apply it to the labor market. An alternative approach is market power due to search or information frictions (Menzio (2024)), which seems less plausible in the rural economy, where information flows are not very frictional. Finally, there is the approach of labor market power due to comparative advantage over jobs (Bils et al. (2025)), which also seems somewhat less plausible in a setting where jobs are not very skill-intensive. This paper presents the results of workfare expansion in one of these three settings. Yet another source of market power could be collusion between employers; and there is some industry-specific evidence in the development literature to that extent. The data in this setting cannot inform the presence or absence of collusion; and the model will rest on the assumption of oligopsonistic competition. The limited point made in this paper, thus, is as follows: the market shares mechanism embedded in a particular class of macroeconomic models predicts that a 30% expansion in the size of workfare generates a 1.1% to 3.9% increase in market wages and 1.4% to 4.3% increase in market employment.

The theoretical insight extends beyond workfare. There is large cross-country variation in the share of the laborforce employed by the government (13% in the US, 30% in Europe). The variation is amplified at finer geography cross occupation resolutions. Public employment takes away a share of the local labor market; increasing the elasticity of labor supply facing firms and reducing their ability to mark down wages. Even a completely unproductive public sector, thus, can deliver some efficiency gains. This insight speaks to the recent on the size of the public sector in the US.

This paper relates to recent literatures on labor market power in macroeconomics and development. It also relates to the literature that uses structural macroeconomic models to interpret and extend on insights from randomized experiments (Buera et al. (2020), Kaboski and Townsend (2011)).

I build on the model of heterogenous firms with endogenous markdowns developed in Berger et al. (2022), Berger et al. (2025). These papers take an Atkeson and Burstein (2008) style model of product market power and apply it to the labor market. This paper extends on the aggregation structure developed in Berger et al. (2022) by introducing public employment as an additional level in the nested CES labor supply disutility. The quantitative excercise involves computing a version of the model that is not block recursive; i.e. the within-market markdowns are not independent of aggregates. Berger et al. (2025) discuss *spillovers* of a firm's wage policies on other firms. This paper studies spillovers of public employment on the market economy.

I contribute to the literature on labor market power in macro-development. Amodio et al. (2025), study the role of self-employment options in attenuating firm labor market power. Brooks et al. (2021) quantify the aggregate reduction in labor share of income attributable to wage markdowns in China and India. Armangué-Jubert et al. (2025) undertake a cross-country analysis to study how labor market competitiveness varies with GDP per capita. This paper seeks to quantify the role of workfare programs in attenuating losses from market power.

Finally, my paper speaks to the largely empirical literature on workfare programs (Imbert and Papp (2020), Imbert and Papp (2015), Klonner and Oldiges (2022)). Besley and Coate (1992) provide a theory of employment guarantees which is concerned with the self-selection feature. I directly build on Muralidharan et al. (2023) who undertake a randomized experiment to to study the equilibrium effects of workfare on the market economy.

2. Evidence

India's National Rural Employment Guarantee Scheme (NREGS) was launched in 2007. Under this program, every rural citizen is entitled to upto a 100 days of employment on public works, at a state mandated wage, available on demand. Wages are set by sub-state boards, and are usually set to be approximately the same as the going market wage.

Due to constraints on state capacity, the program is subject to a number of implementation frictions, so that the employment guarantee implicit in the program is not very credible (cite). Muralidharan et al. (2023) improve a particular implementation friction in a random set of districts in the Indian state of Andhra Pradesh. The improvement consists of replacing erstwhile cash disbursal of program payments with digital payments. Digital payments improve program access for a number of reasons; the two main reasons they point out are: (1) reduced corruption leakages improve the efficiency of the program and (2) reduced delays in payment arrivals, which are a substantial program access friction in this setting.

In line with the improvement in program access, employment in the workfare program increases by 30%. Employment in the market ('private') economy, excluding the workfare sector, increases by 18%. Wages in the market economy also increase by 10%. These effects are not estimated very precisely; the employment effect is significant at 10% and the 90% confidence interval around it is [1%,34%]. The wage effect is significant at 5% and the 95% confidence interval around it is [2%,19%]. Employment effects are bigger in villages with higher HHI, where HHI is measured using the concentration of landholdings and is interpreted as proxying for labor market concentration. The employment and wage effects are estimated for the month of June, when demand for agricultural labor is low and hence the demand for workfare jobs is high. There is a substantial increase in household spending and uptake of new loans in treated districts. There is also some evidence of entry of new non-agricultural firms. Finally, Muralidharan et al. (2023) have data on the value of public assets created under the program, and they find it to be very small.

Multiple mechanisms operating in the background complicate the interpretation of the wage and employment treatment effects. Household demand increases and financial frictions reduce; which could both drive an increase in employment and wages even with competitive labor markets. I evaluate the quantitative importance of the labor market power channel.

3. Model

3.1. Environment

Suppose there are $j \in [0, 1]$ villages; within each village there is a market ('private') sector and a workfare program. The market sector consists of m_j firms heterogenous in productivity. The workfare sector consists of a workfare project with exogenously given wage $w_{m_j j} = \bar{w}_j$. A representative household chooses to send measure n_{ij} workers to firm i in village j and a measure $\{n_j^w\}$ workers to the workfare program. The disutility from labor is nested CES; the household imperfectly substitutes between firms within a village, between the market and workfare sector within a village and across

villages. The household's problem is stated as:

$$\max_{\{c_{ij}, n_{ij}, n_j^w\}} U(C, N)$$

$$\text{s.t.} \quad C = \int_0^1 \left(\sum_i w_{ij} n_{ij} + \bar{w}_j n_j^w \right) dj + (1-\tau)\Pi$$

$$\text{where} \quad C = \int_0^1 \sum_i c_{ij} dj$$

$$\text{and} \quad n_j^p = \left(\sum_{i=1}^{m_j} (n_{ij})^{\frac{\eta+1}{\eta}} \right)^{\frac{\eta}{\eta+1}}$$

$$\text{and} \quad n_j = \left((n_j^p)^{\frac{\gamma+1}{\gamma}} + \delta (n_j^w)^{\frac{\gamma+1}{\gamma}} \right)^{\frac{\gamma}{\gamma+1}}$$

$$\text{and} \quad N = \left(\int_0^1 n_j^{\frac{\theta+1}{\theta}} \, dj \right)^{\frac{\theta}{\theta+1}}$$

The output of all firms is perfectly substitutable in the household's utility, and the price of the good is normalized to 1. The household solves a static problem, and hence consumes its entire income. The household's labor income consists of payments from all firms i in village j. We assume that the household owns all the firms, so that profits are rebated back to the household subject to a tax τ that is used to finance the workfare program. The household allocates labor across villages; thereafter, between private and workfare sectors within a village and finally, across firms within the private sector in each village. The CES structure implies that the household experiences convexly increasing disutility from sending workers to a particular firm; thus, to grow big, the firm has to compensate the household with a higher wage (Berger et al. (2022)). The term δ regulates the relative disutility of sending labor to workfare instead of private; when δ is large the relative disamenity associated with workfare is large. Varying δ enables us to vary the share of the local labor market that goes to workfare. Note also that in this model, workfare is completely unproductive. The workfare sector only matters for the economy through its effects on the household's allocation of labor.

The household's first order condition implies the labor supply curve facing firm i in village j:

$$w_{ij} = \left(\frac{n_{ij}}{n_j^p} \right)^{\frac{1}{\eta}} \left(\frac{n_j^p}{n_j} \right)^{\frac{1}{\gamma}} \left(\frac{n_j}{N} \right)^{\frac{1}{\theta}} W \tag{1}$$

where the following are true by definition

$$w_j^p n_j^p = \sum_{i=1}^{m_j} w_{ij} n_{ij}$$

$$w_j n_j = w_j^p n_j^p + \bar{w}_j n_j^w$$

$$WN = \int_0^1 w_j n_j \, dj$$

which, given the CES structure imply wage indices:

$$w_j^p = \left(\sum_i^{m_j} w_{ij}^{1+\eta}\right)^{\frac{1}{1+\eta}}$$
$$w_j = \left((w_j^p)^{1+\gamma} + \left(\frac{1}{\delta^\gamma}\right)(\bar{w}_j)^{1+\gamma}\right)^{\frac{1}{1+\gamma}}$$
$$W = \left(\int_0^1 w_j^{1+\theta} dj\right)^{\frac{1}{1+\theta}}$$

Firms draw idiosyncratic productivity z_{ij} and operate a decreasing returns to scale technology. The firm's problem is given by:

$$\max_{n_{ij}} \left\{ z_{ij} n_{ij}^\alpha - w_{ij}\left(n_{ij}, n_j^p, n_j^w, N, W\right) n_{ij} \right\}$$
$$\text{subject to} \quad w_{ij}\left(n_{ij}, n_j^p, n_j^w, N, W\right) = \left(\frac{n_{ij}}{n_j^p}\right)^{\frac{1}{\eta}} \left(\frac{n_j^p}{n_j}\right)^{\frac{1}{\gamma}} \left(\frac{n_j}{N}\right)^{\frac{1}{\theta}} W$$

where the firm faces an upward sloping labor supply given by (1).

The firm's first order condition is:

$$\alpha \frac{y_{ij}}{n_{ij}} = w_{ij} + \frac{\partial w_{ij}}{\partial n_{ij}} n_{ij}$$

Define $\varepsilon_{ij} := \left[\frac{\partial log w_{ij}}{\partial log n_{ij}}\right]^{-1}$. Then,

$$w_{ij} = \mu_{ij} \alpha \frac{y_{ij}}{n_{ij}}, \quad \mu_{ij} = \frac{\varepsilon_{ij}}{1+\varepsilon_{ij}} \tag{2}$$

where μ_{ij} is the wage markdown induced by market power. We differentiate (1) to obtain:

$$\begin{aligned} \frac{\partial log w_{ij}}{\partial log n_{ij}} &= \frac{1}{\eta} + \frac{\partial log n_j^p}{\partial log n_{ij}}\left(\frac{1}{\gamma} - \frac{1}{\eta}\right) + \frac{\partial log n_j}{\partial log n_{ij}}\left(\frac{1}{\theta} - \frac{1}{\gamma}\right) \\ &= \frac{1}{\eta} + s_{ij}\left(\frac{1}{\gamma} - \frac{1}{\eta}\right) + \left(\frac{n_j^p}{n_j}\right)^{\frac{\gamma+1}{\gamma}} s_{ij}\left(\frac{1}{\theta} - \frac{1}{\gamma}\right) \end{aligned} \tag{3}$$

where s_{ij} is the payroll share given by:

$$s_{ij} = \frac{n_{ij} w_{ij}}{\sum_i^{m_j} n_{ij} w_{ij}}$$

Definition 1. *(Equilibrium) Given $\{\bar{w}_j\}$, an equilibrium consists of $\left\{w_{ij}, n_{ij}, n_j^w\right\}_{\forall i,j}$, household consumption C, output Y, government spending G and a tax rate τ such that*

1. *each $\left\{w_{ij}, n_{ij}, n_j^w\right\}$ consistent with the household's labor supply curve*
2. *each n_{ij} solves the firm's problem*

3. the government's budget is balanced

$$G = \int_0^1 \bar{w}_j n_j^w dj = WN - W^p N^p$$
$$G = \tau \Pi$$

4. the goods market clears

$$C = WN + (1-\tau)\Pi$$
$$C = WN + Y - W^p N^p - \tau \Pi$$
$$C = Y$$

Note that in this model, labor market clearing is at the firm level.

3.2. Results

Let x_j collect terms in the inverted labor supply curve that are common to all firms in market j. The following equations are immediate:

$$mrpl_{ij} = \alpha z_{ij} n_{ij}^{\alpha-1}$$
$$n_{ij} = w_{ij}^{\eta} x_j$$
$$w_{ij} = \mu_{ij} mrpl_{ij}$$

These together with (1) and (3) imply:

Proposition 1. *(Within-village markdowns) Markdowns in market j, $\{\mu_{ij}\}_{i=1}^{m_j}$ are a solution to the following system:*

$$s_{ij} = \frac{[\mu(s_{ij}) z_{ij}]^{\frac{\eta+1}{1+\eta(1-\alpha)}}}{\sum_i^{m_j} [\mu(s_{ij}) z_{ij}]^{\frac{\eta+1}{1+\eta(1-\alpha)}}} \tag{4}$$

$$\mu(s_{ij}) = \frac{\varepsilon(s_{ij})}{\varepsilon(s_{ij}) + 1} \tag{5}$$

$$\varepsilon(s_{ij}) = \left((1-s_{ij}) \frac{1}{\eta} + s_{ij} \left(1 - \left(\frac{n_j^p}{n_j} \right)^{\frac{\gamma+1}{\gamma}} \right) \frac{1}{\gamma} + s_{ij} \left(\frac{n_j^p}{n_j} \right)^{\frac{\gamma+1}{\gamma}} \frac{1}{\theta} \right)^{-1} \tag{6}$$

Unlike Berger et al. (2022) equilibrium markdowns cannot be recovered by solving (4), (5), (6) as a fixed point system in shares. This requires a guess of the market-level outcome $\frac{n_j^p}{n_j}$. Under the model structure developed here, the household does not arrive at a decision node in which it directly chooses the allocation of private labor across villages. The household chooses the allocation of labor across villages, and the allocation across private and workfare labor within a village. Consequently, CES-type private labor and wage indices, that only aggregate over private firms across villages do not exist. Aggregation is particularly convenient in a model which admits such CES-type indices. We assume that workfare wage in market j is proportional to the equilibrium private wage in market j. We will use this assumption to build intuition about the model mechanism. For the purpose of quantititive analysis, this assumption will be relaxed.

Assumption 1. $\bar{w}_j = \phi w_j^p$

Under Assumption 1, we have two results.

Proposition 2. *(Block-recursivity) Under assumption 1, markdowns in market j, $\{\mu_{ij}\}_{i=1}^{m_j}$, can be recovered without knowing aggregates:*

$$
\begin{aligned}
s_{ij} &= \frac{[\mu(s_{ij})z_{ij}]^{\frac{\eta+1}{1+\eta(1-\alpha)}}}{\sum_i^{m_j}[\mu(s_{ij})z_{ij}]^{\frac{\eta+1}{1+\eta(1-\alpha)}}} \\
\mu(s_{ij}) &= \frac{\varepsilon(s_{ij})}{\varepsilon(s_{ij})+1} \\
\varepsilon(s_{ij}) &= \left((1-s_{ij})\frac{1}{\eta} + s_{ij}(1-\Phi^{-(1+\gamma)})\frac{1}{\gamma} + s_{ij}\Phi^{-(1+\gamma)}\frac{1}{\theta}\right)^{-1}
\end{aligned} \tag{7}
$$

where

$$
\Phi = \left(1 + \frac{\phi^{\gamma+1}}{\delta^{\gamma}}\right)^{\frac{1}{1+\gamma}} \tag{8}
$$

where Φ is an index of workfare perfomance, and $\Phi^{-\gamma}$ is the private market share of the village.

Consider the case of symmetric firms; without heterogeniety, $s = \frac{1}{m_j}$. As workfare improves, (δ goes down), Φ goes up, and so in the inverse elasticity expression, the weight on $\frac{1}{\gamma}$ goes up relative to the weight on $\frac{1}{\theta}$. Provided workfare and market jobs are more substitutable than jobs across villages, workfare expansions increase the inverse elasticity of labor supply, driving μ_{ij} to increase. Thus, workfare expansions reduce the inefficiency due to labor market power. This is true regardless of adjustments on the market employment and wage margins.

Corollary 1. *When $\gamma > \theta$, workfare expansions reduce the inefficiency due to labor market power.*

We now define the aggregate private employment index. This index aggregates over the private labor across villages, i.e., excluding labor at the workfare sector in each village.

$$
N^p = \left(\int_0^1 \left(n_j^p\right)^{\frac{\theta+1}{\theta}}\right)^{\frac{\theta}{\theta+1}} \tag{9}
$$

Under Assumption 1, we can consistently derive a CES-type wage index associated with (9). That is, there exists a number W^p, such that

$$
W^p N^p = \int_0^1 w_j^p n_j^p dj
$$

and W^p has the expected form,

$$
W^p = \left(\int_0^1 \left(w_j^p\right)^{1+\theta}\right)^{\frac{1}{1+\theta}}
$$

Given equilibrium markdowns in the continuum of villages and firm productivities, we have everything needed to solve for aggregates.

PROPOSITION 3. *(Aggregation) The following system of six equations pins down aggregates* $\{Y, C, W^p, N^p, W, N\}$.

$$Y = \Omega \tilde{Z} \left(N^p\right)^{\alpha} \qquad W^p = \alpha \frac{\mu}{\Omega} \frac{Y}{N^p} \qquad W = \frac{-U_N}{U_C} \tag{10}$$

$$W = \Phi W^p \qquad N = \Phi^{\gamma} N^p \qquad C = Y \tag{11}$$

where the constants $\tilde{Z}$, Ω *and* μ *are;*

$$Z = \left[\int_0^1 z_j^{\frac{1+\theta}{1+\theta(1-\alpha)}} dj\right]^{\frac{1+\theta(1-\alpha)}{1+\theta}}, \quad z_j = \left[\sum_{i=1}^{m_j} z_{ij}^{\frac{1+\eta}{1+\eta(1-\alpha)}}\right]^{\frac{1+\eta(1-\alpha)}{1+\eta}}$$

$$\mu = \left[\int_0^1 \left(\frac{z_j}{Z}\right)^{\frac{1+\theta}{1+\theta(1-\alpha)}} \mu_j^{\frac{1+\theta}{1+\theta(1-\alpha)}} dj\right]^{\frac{1+\theta(1-\alpha)}{1+\theta}}, \quad \mu_j = \left[\sum_{i=1}^{m_j} \left(\frac{z_{ij}}{z_j}\right)^{\frac{1+\eta}{1+\eta(1-\alpha)}} \mu_{ij}^{\frac{1+\eta}{1+\eta(1-\alpha)}}\right]^{\frac{1+\eta(1-\alpha)}{1+\eta}}$$

$$\Omega = \int_0^1 \left(\frac{z_j}{Z}\right)^{\frac{1+\theta}{1+\theta(1-\alpha)}} \left(\frac{\mu_j}{\mu}\right)^{\frac{\alpha\theta}{1+\theta(1-\alpha)}} \omega_j dj, \quad \omega_j = \sum_{i=1}^{m_j} \left(\frac{z_{ij}}{z_j}\right)^{\frac{1+\eta}{1+\eta(1-\alpha)}} \left(\frac{\mu_{ij}}{\mu_j}\right)^{\frac{\eta\alpha}{1+\eta(1-\alpha)}}$$

In (10), the first equation is an aggregate production function, the second is the 'representative' firm's labor first order condition and the third equation is the representative household's labor first order condition. In (11), the first two equations represent the aggregate wage and labor indices W and N in terms of the aggregate private wage and labor indices W^p and N^p. These relations are due to assumption 1 which represents the workfare wage as a constant scaling of the equilibrium private wage. The third equation in (11) is what will be shortly introduced (in definition 1) as the goods market clearing condition. $\tilde{Z}$, Ω and μ are respectively indices of aggregate productivity, a misallocation wedge that comes from the covariance between markdowns and productivity and an aggregate markdown that occurs due to market power. In Berger et al. (2022) $W = W^p$ and $N = N^p$ so that (10) and (11) reduce to four equations in four unknowns.

In what follows, household preferences are parametrized to be CRRA; $\frac{1}{\nu}$ is the Frisch elasticity of labor supply.

$$U(C, N) = \frac{C^{1-\sigma}}{1-\sigma} - \frac{N^{1+\nu}}{1+\nu} \tag{12}$$

3.2.1. Employment and wage margins.

Suppose each village consists of a single firm with productivity z and workfare. Using proposition 3, equilibrium firm-level output y, employment n, wage w, markdown μ are a solution to the following system of equations:

$$y = zn^{\alpha} \qquad w = \frac{\mu y}{n} \qquad \Phi^{1-\gamma\nu} w = n^{\nu} y^{\sigma} \qquad \mu = \frac{1}{1+\varepsilon^{-1}} \qquad \varepsilon^{-1} = \left(\frac{1}{\gamma} + \Phi^{-(1+\gamma)}\left(\frac{1}{\theta} - \frac{1}{\gamma}\right)\right)$$

This system yields a solution for (n, w) (derivations in appendix). We begin by obtaining firm employment and wage responses to an increase in the index of workfare performance Φ in the competitive benchmark. Here, we force the firm to equate the wage to the marginal product of labor so that $\mu = 1$.

Result 1. *(Competitive benchmark) In the model with a single firm, workfare and no market power inefficiency,*

$$\frac{d\log n}{d\log\Phi} = (1-\gamma\nu)\frac{1}{\alpha(\sigma-1)+\nu+1}$$
$$\frac{d\log w}{d\log\Phi} = (\gamma\nu-1)\frac{1-\alpha}{\alpha(\sigma-1)+\nu+1}$$

Result 1 demonstrates that in the competitive benchmark, workfare expansions always drive firm wage and employment in opposite directions. This is intuitive, since in the competitive benchmark, workfare expansions only generate shifts in the labor supply curve facing the firm. The sign of the responses depends on the size of the Frisch elasticity of labor supply relative to the size of the substitutability of market and workfare jobs. If market and workfare jobs have an elasticity of substitution that is larger than the Frisch elasticity of labor supply, workfare expansions reallocate labor from the market economy to the workfare sector. This shifts labor supply inward, increasing the market wage and reducing market employment. On the other hand, if the elasticity of substitution is smaller than the Frisch elasticity of labor supply, workfare expansions drive a net inflow of labor into the economy. This increases the supply of labor to both, the workfare and market sectors; increasing firm employment and reducing firm wage.

In the economy featuring markdown inefficiency, workfare expansions have an additional effect on the size of the markdown.

Result 2. *(Markdown inefficiency) In the model with a single firm, workfare and the markdown inefficiency, In the model featuring markdown inefficiency ($\mu < 1$),*

$$\frac{d\log n}{d\log\Phi} = (1-\gamma\nu)\frac{1}{\alpha(\sigma-1)+\nu+1} + \frac{1}{\alpha(\sigma-1)+\nu+1}\frac{dlog\mu}{dlog\Phi}$$
$$\frac{d\log w}{d\log\Phi} = (\gamma\nu-1)\frac{1-\alpha}{\alpha(\sigma-1)+\nu+1} + \frac{\alpha\sigma+\nu}{\alpha(\sigma-1)+\nu+1}\frac{dlog\mu}{dlog\Phi}$$

Result 2 demonstrates that in the model with inefficiency, workfare expansions can increase firm employment and wages provided the markdown reduction is big enough. The intuition is that in the model with inefficiency, workfare expansions not only shift the labor supply curve, but also make it flatter. The markdown effect captures the latter force which is absent in the competitive benchmark. Result to enables a characterization of the values of γ and ν that admit positive employment and wage responses to workfare expansions. Figure 1 demonstrates. In this excercise, $\sigma = 2, \alpha = 0.8, \theta = 0.45, \phi = 1, \delta = 1$.

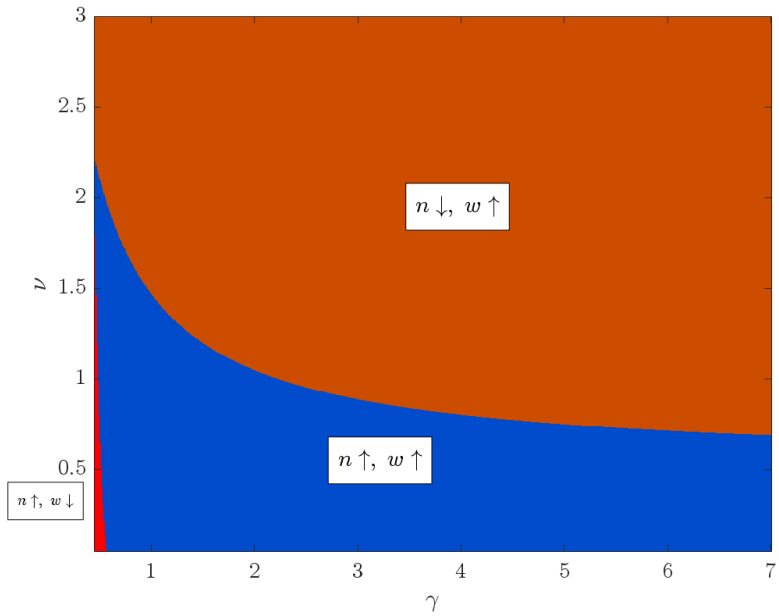
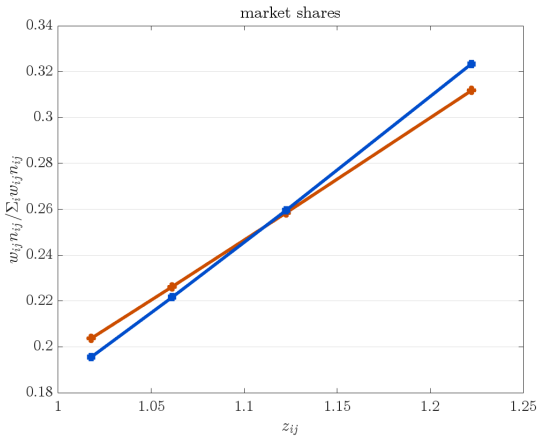


Figure 1. Parameters that admit increase in market employment and wages

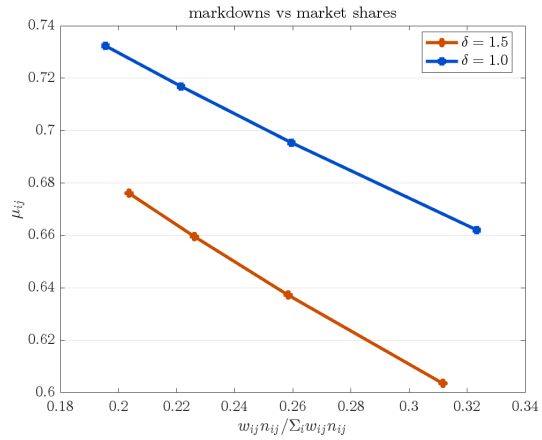
For large values of γ, a given improvement in workfare implies a large markdown reduction. But it also implies a large reallocation of labor away from workfare. Consequently, the household needs to increase aggregate labor supply following the workfare expansion sufficiently to offset the reallocation effect. Thus, the inverse Frisch, ν needs to be small when γ to achieve an increase in market employment.

3.2.2. Firm size, market power and misallocation.

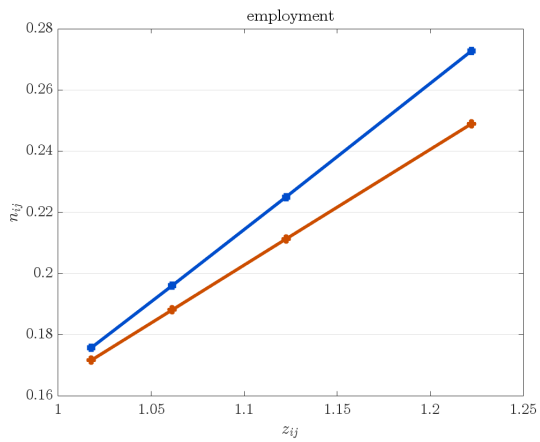
We introduce a simple form of heterogeniety; each village contains four firms with productivities fixed at the 10th, 30th, 50th and 70th percentiles of the Pareto distribution. The red line corresponds to the economy with inefficient workfare ($\delta = 1.5$) and the blue line corresponds to the economy with improved workfare ($\delta = 1$). Figure 2, panel 1 shows that in the economy with improved workfare, productive firms become bigger in the local labor market relative to unproductive firms. Panel 2 shows that workfare improvements reduce the wage markdown at all firms; but the size of the reduction is larger for big firms. Thus, the workfare improvement flattens the gradient of the relationship between shares and markdowns. What explains the reallocative effects of workfare? The inverse elasticity expression (7) shows that as $s \to 0$, i.e. the firm is small in the local labor market, workfare improvements do not affect the elasticity of the labor supply curve facing it. Intuitively, when the firm is small, it is primarily competing with other firms in the village. On the other hand, when the firm is big in the local labor market, i.e. $s \to 1$, it is competing for labor with workfare and with other villages; hence, workfare improvements have bigger impacts on markdowns at large firms.



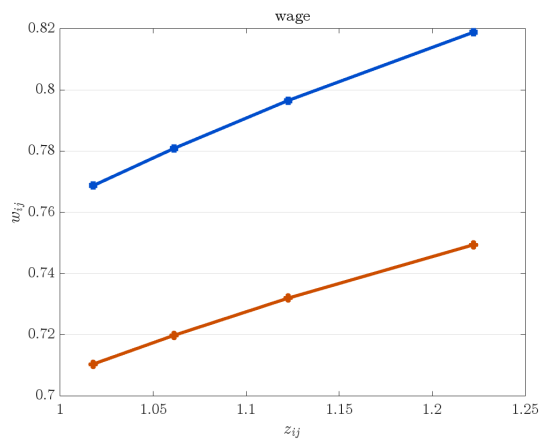
(a) Panel (a): Shares vs markdowns



(b) Panel (b): Wages



(c) Panel (c): Employment



(d) Panel (d): Shares

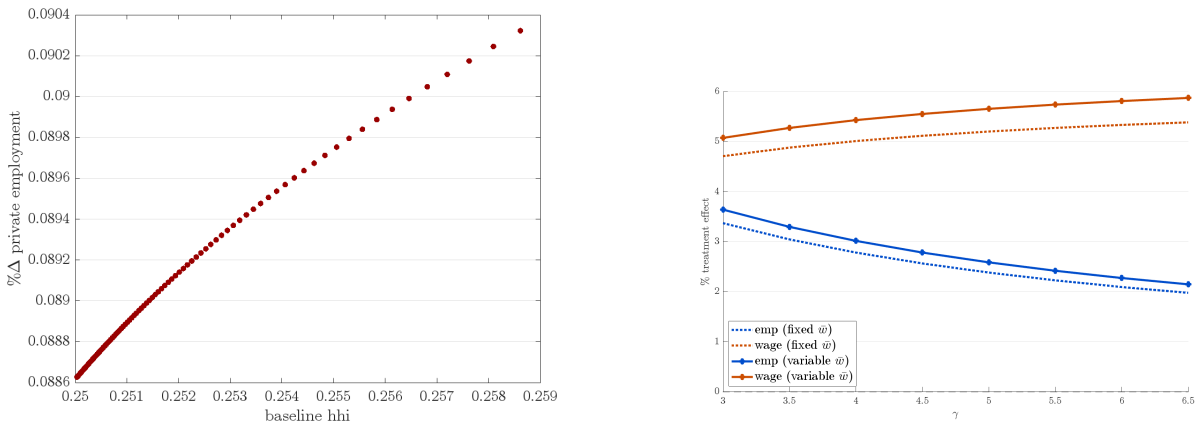
Figure 2. Moving $\delta = 1.5 \rightarrow \delta = 1$

While the big, producitive firms' payroll becomes a bigger share of the local aggregate payroll, relative adjustments on the wage and employment margins at large relative to small firms are ambiguous. In this particular example, panels 3 and 4 show that employment and wages go up at all firms, but by more at the most productive firms. Since with Pareto productivity draws, high payroll concentration markets are markets with high producitivity draws, the size of the employment effect is bigger in more concentrated markets (Figure 3, panel a).

3.2.3. Quantitative importance of workfare.

The assumption $\bar{w}_j = \phi w_j^p$ is not innocuous; workfare wages endogenously increase as the equilibrium market wage increases. Relaxing this assumption loses us analytical tractability. Further, makrdowns within a local labor market now depend on aggregates. To compute the equilibrium, we guess the village-specific endogenous wedge $\phi_j = \frac{\bar{w}_j}{w_j^p}$ and then iterate on the guess using the structure of the model (computation algorithm detailed in appendix). In Figure 3 panel b, we

target an initial workfare-market labor ratio of 0.5 and implement an increase in the size of workfare of 30%. With thick lines, we show the market employment and wage effects in the endogenous workfare wage model. With dotted lines, we show the market employment and wage effects when the workfare improvement is implemented by fixing the workfare wage to the baseline steady state equilibrium private wage (without allowing endogenous adjustment). We observe that removing the endogenous feedback marginally attenuates the size of employment and wage effects.



(a) Workfare in more concentrated markets

(b) Quantitative effects of workfare

Figure 3

4. Quantitative experiment

I present an elementary calibration of the model to evaluate the quantitative importance of the labor market power channel. Berger et al. (2022) exploit state-level corporate tax changes to estimate labor supply elasticities at the firm level, using longitudinal census data. Thereafter, they estimate elasticities of substitution across firms and labor markets to target the empirical relationship between payroll shares and labor supply elasticities. Here, we take a different approach. We ask, what is the biggest market employment and wage effect plausible in a calibration? There is a set of parameters that can be informed by empirical moments; for three key parameters (inverse Frisch, substitutability of jobs across firms and substitutability of jobs across market and workfare) there is no guidance in the data. These parameters will be chosen to generate the maximum plausible market employment and wage effects in the model. The strategy is:

1. Classify parameters into three categories: $A = \{\sigma, \xi\}$[1], $B = \{\nu, \eta, \gamma\}, C = \{\phi, \delta, m_j, \alpha, \theta, \Delta\delta\}$
2. Given A and B, pick C to match 6 moments
3. Choose B to maximize market employment and wage treatment effects

Treatment effects are computed as $\int_0^1 \frac{y^p_{j,post} - y^p_{j,pre}}{y^p_{j,pre}} dj$. We first solve the model before workfare improvement, and match the mean HHI, workfare relative to market wage ($\int_0^1 \frac{\bar{w}_{j,pre}}{w^p_{j,pre}} dj$), workfare

[1]We hope to use data on the landholdings distribution to inform the dispersion parameter ξ, but that data is not publicly accesible, so we externally fix ξ in this calibration.

relative to market employment ($\int_0^1 \frac{n^w_{j,pre}}{n^p_{j,pre}} dj$), the labor share ($\mu\alpha$) and the wage markdown (μ). Thereafter, we fix the workfare wage to the level in this 'baseline' ($w^p_{j,pre}$), and then reduce δ to match a 30% expansion in the size of workfare labor.

The markdown estimate used in the calibration, $\mu = 0.72$, is Brooks et al. (2021)'s upper bound of wage markdown estimates for Indian manufacturing. Given that the setting for Muralidharan et al. (2023)'s experiment is rural, there may be an important role for agriculture. We do another calibration that targets Brooks et al. (2021)'s lower bound of markdown estimates $\mu = 0.30$. This calibration requires reducing the number of firms so we do not hit the mean HHI of 0.03 in Muralidharan et al. (2023)'s baseline data. The mean HHI is computed on the basis of landholdings census data. To the extent that not all households are employers of labor, HHI derived from the census data may not be a good measure labor market concentration. However, in this calibration, we struggle to match the labor share of 0.52. Consequently, we do yet another calibration to target Armangué-Jubert et al. (2025)'s markdown estimate of $\mu = 0.54$ for India (this is derived using the World Bank Enterprise Survey).

The baseline calibration ($\mu = 0.72$, mean HHI = 0.03) delivers that a 30% expansion in workfare generates a 1.4% increase in market employment and a 1.3% increase in the market wage. It predicts a coefficient on the interaction between treatment and payroll HHI relative to baseline mean private employment of 0.22. 1.4% employment effect compares to the point estimate of 18% in the data; and the confidence interval around it is [1.06%, 34.38%]. 1.3% wage effect compares with a point estimate of 10% in the data, and the confidence interval around it is [1.50%, 18.82%]. Finally the treatment gradient in HHI of 22% compares with a point estimate of 58% in the data, and the confidence interval around it is [1.1%,115.3%].

The Armangué-Jubert et al. (2025) calibration ($\mu = 0.54$) delivers bigger effects: a 30% expansion in workfare generates 2.5% increase in market employment and 2.8% increase in wage. The treatment gradient in HHI however falls to 1.6%. The calibration that targets Brooks et al. (2021)'s lower bound markdown estimate of $\mu = 0.30$ does not get the labor share of 0.52 right[2]. Nonetheless, we report that we find higher effects: a 30% expansion in workfare generates a 4.3% increase in employment and a 3.9% increase in the wage. It delivers a treatment gradient in HHI of 2.4%.

4.1. Discussion

A model that matches the initial workfare-private labor share, the initial workfare-private relative wage, the markdown estimate and the labor share delivers up to 2.5% increase in market employment and 2.8% increase in the market wage in response to a 30% expansion in the size of the workfare sector. These numbers are within the confidence intervals around estimates in Muralidharan et al. (2023). Calibrations that feature more labor market power generate larger employment and wage effects. Calibrations that feature fewer firms in each village generate larger employment and wage effects. Most importantly, a larger than 30% workfare expansion in the model would deliver larger employment and wage effects. Thus, through the lens of this model, a 30% workfare expansion justifies < half of the aggregate wage effect < one third of the aggregate employment effect. Through the lens of this model, employment and wage effects of workfare expansions are not quantitatively first order. Nonetheless, these estimates help place a bound on how productive the workfare sector needs to be to justify workfare-based redistribution.

[2]The labor share equals the DRS parameter α times the markdown μ. With a labor share of 0.52 and markdown of 0.30, the implied α is > 1.

Parameters	Interpretation	Targets	Source
Assigned			
$\boldsymbol{\sigma}$	curvature in consumption	log utility	
$\boldsymbol{\xi}$	Pareto shape		
Calibrated			
$\boldsymbol{\nu}$	inverse Frish		
$\boldsymbol{\eta}$	substitution across firms		
$\boldsymbol{\gamma}$	substitution workfare vs private		
$\boldsymbol{m_j = m}$	number of firms	mean HHI (0.030)	Muralidharan et al (2022)
$\boldsymbol{\phi}$	$\bar{w}_{j,baseline} = \phi w^p_{j,baseline}$	workfare/market wage (1)	Muralidharan et al (2022)
$\boldsymbol{\delta}$	workfare disutility shifter	workfare/market labor (0.57)	Muralidharan et al (2022)
$\boldsymbol{\alpha}$	exponent on n	labor share (0.52)	Jat and Ramaswami (2024)
$\boldsymbol{\theta}$	substitution across villages	wage markdown ($\mu = 0.72$)	Soundararajan et al (2024), Brooks et al (2021)
$\boldsymbol{\Delta\delta}$	program improvement	workfare treat effect (0.30)	Muralidharan et al (2022)

Table 1. Calibration

	Interpretation / Moment	Model	Data
Targeted Moments			
workfare/market wage ratio		1	1
workfare/market labor ratio		0.57	0.57
mean HHI		0.033	0.03
labor share		0.52	0.52
markdown μ		0.72	0.72
workfare labor treatment effect		0.30	0.30
Parameters			
σ	risk aversion	1	
ξ	Pareto shape	6	
ν	inverse Frisch	0.1	
η	substitution across firms (BHM 6.96)	10	
γ	substitution workfare vs private	10	
m	number of firms	33	
ϕ	workfare/private wage ratio	1	
δ	workfare disutility	1.06	
α	DRS (labor exponent)	0.7	
θ	substitution across villages (BHM 0.45)	0.08	
$\Delta\delta$	workfare improvement	-0.04	
Non-Targeted Moments			
market employment effect		0.014	0.18 (0.01,0.34)
market wage effect		0.013	0.10 (0.02,0.19)
employment effect with ↑ HHI		0.22	0.59 (0.01,1.15)

Note: Employment effect with ↑ HHI is relative to baseline mean employment.

5. Conclusion and next steps

I study workfare in a static model of oligopsonistic competition with endogenously variable wage markdowns. The model has empirically consistent predictions for market employment and wages, and the gradient of effects with respect to labor market concentration. I calibrate the model to match some salient features of the rural Indian economy. A 30% workfare expansion predicts 1.4 to 4.3% increase in market employment and 1.3 to 3.9% increase in market wages. Redistribution typically has efficiency costs; I show a specific case in which workfare-based redistribution could generate efficiency gains.

Going ahead, I want to discover better ways of validating this model. Once I am sufficiently convinced, I want to formulate the problem of a planner choosing between redistribution through unconditional cash transfers and redistribution through workfare in the presence of labor market power.

6. Appendix 1: Derivations

6.1. Market and economy indexes

Taking first order conditions wrt n_{ij} for $j = 1...m_{j-1}$ and n_j^w;

$$-U_N \frac{dN}{dn_j} \frac{dn_j}{dn_j^p} \frac{dn_j^p}{dn_{ij}} = U_C w_{ij}$$
$$-U_N \frac{dN}{dn_j} \frac{dn_j}{dn_j^w} = U_C \bar{w}$$

which simplify to

$$\frac{-U_N}{U_C} N^{\frac{-1}{\theta}} n_j^{\frac{1}{\theta}} n_j^{\frac{-1}{\gamma}} (n_j^p)^{\frac{1+\gamma}{\gamma}} = \sum_i^{m_j} w_{ij} n_{ij} \tag{13}$$
$$\frac{-U_N}{U_C} N^{\frac{-1}{\theta}} n_j^{\frac{1}{\theta}} n_j^{\frac{-1}{\gamma}} (n_j^w)^{\frac{1+\gamma}{\gamma}} \delta = \bar{w} n_j^w \tag{14}$$

Adding up (13) and (14),

$$\frac{-U_N}{U_C} N^{\frac{-1}{\theta}} n_j^{\frac{1}{\theta}} n_j^{\frac{-1}{\gamma}} n_j^{\frac{\gamma+1}{\gamma}} = w_j^p n_j^p + \bar{w} n_j^w \tag{15}$$
$$\frac{-U_N}{U_C} N^{\frac{-1}{\theta}} n_j^{\frac{1+\theta}{\theta}} = w_j n_j \tag{16}$$
$$\frac{-U_N}{U_C} N^{\frac{-1}{\theta}} \int_0^1 n_j^{\frac{1+\theta}{\theta}} dj = \int_0^1 w_j n_j dj$$
$$\frac{-U_N}{U_C} N^{\frac{-1}{\theta}} N^{\frac{\theta+1}{\theta}} = WN$$
$$\frac{-U_N}{U_C} = W \tag{17}$$

Substituting (17) into (13),

$$w_{ij} = \left(\frac{n_{ij}}{n_j^p}\right)^{\frac{1}{\eta}} \left(\frac{n_j^p}{n_j}\right)^{\frac{1}{\gamma}} \left(\frac{n_j}{N}\right)^{\frac{1}{\theta}} W$$

We derive some relations that will help in the following derivations. Using (16),

$$\begin{aligned} WN^{\frac{-1}{\theta}} n_j^{\frac{1}{\theta}} &= w_j \\ n_j &= \left(\frac{w_j}{W}\right)^{\theta} N \end{aligned} \tag{18}$$

Using (13),

$$\begin{aligned} WN^{\frac{-1}{\theta}} n_j^{\frac{1}{\theta}} n_j^{\frac{-1}{\gamma}} (n_j^p)^{\frac{1+\gamma}{\gamma}} &= w_j^p n_j^p \\ W\left(\frac{n_j}{N}\right)^{\frac{1}{\theta}} \left(\frac{n_j^p}{n_j}\right)^{\frac{1}{\gamma}} &= w_j^p \\ n_j^p &= \left(\frac{w_j^p}{W}\right)^{\gamma} \left(\frac{N}{n_j}\right)^{\frac{\gamma}{\theta}} n_j \end{aligned} \tag{19}$$

Using (14),

$$\begin{aligned} WN^{\frac{-1}{\theta}} n_j^{\frac{1}{\theta}} n_j^{\frac{-1}{\gamma}} (n_j^w)^{\frac{1}{\gamma}} \delta &= \bar{w} \\ n_j^w &= \left(\frac{\bar{w}}{\delta W}\right)^{\gamma} N^{\frac{\gamma}{\theta}} n_j^{\frac{\theta-\gamma}{\theta}} \end{aligned} \tag{20}$$

Using (19),

$$\begin{aligned} n_j^p &= \left(\frac{w_j^p}{W}\right)^{\gamma} N^{\frac{\gamma}{\theta}} n_j^{1-\frac{\gamma}{\theta}} \\ n_j &= \left[\left(\frac{W}{w_j^p}\right)^{\gamma} N^{\frac{-\gamma}{\theta}} n_j^p\right]^{\frac{\theta}{\theta-\gamma}} \end{aligned} \tag{21}$$

Substituting into the first order condition with respect to n_{ij},

$$
\begin{aligned}
WN^{\frac{-1}{\theta}}\left(\left[\left(\frac{W}{w_j^p}\right)^{\gamma} N^{\frac{-\gamma}{\theta}} n_j^p\right]^{\frac{\theta}{\theta-\gamma}}\right)^{\frac{1}{\theta}-\frac{1}{\gamma}} (n_j^p)^{\frac{1}{\gamma}-\frac{1}{\eta}} n_{ij}^{\frac{1}{\eta}} &= w_{ij} \\
WN^{\frac{-1}{\theta}}\left(\left(\frac{W}{w_j^p}\right)^{\gamma} N^{\frac{-\gamma}{\theta}} n_j^p\right)^{\frac{-1}{\gamma}} (n_j^p)^{\frac{1}{\gamma}-\frac{1}{\eta}} n_{ij}^{\frac{1}{\eta}} &= w_{ij} \\
WN^{\frac{-1}{\theta}}\left(\frac{w_j^p}{W}\right) N^{\frac{1}{\theta}} (n_j^p)^{\frac{1}{\gamma}-\frac{1}{\eta}} n_{ij}^{\frac{1}{\eta}} &= w_{ij} \\
\left(\frac{w_{ij}}{w_j^p}\right)^{\eta} n_j^p &= n_{ij}
\end{aligned}
\tag{22}
$$

We can now derive the private sector wage index. Using (22),

$$
\begin{aligned}
w_j^p n_j^p &= \sum_i^{m_j} w_{ij} n_{ij} \\
&= \sum_i^{m_j} w_{ij} \left(\frac{w_{ij}}{w_j^p}\right)^{\eta} n_j^p \\
w_j^p &= \left(\sum_i^{m_j} w_{ij}^{\eta+1}\right)^{\frac{1}{1+\eta}}
\end{aligned}
\tag{23}
$$

Using (19) and (20),

$$
\begin{aligned}
w_j n_j &= w_j^p n_j^p + \bar{w} n_j^w \\
&= w_j^p \left(\frac{w_j^p}{W}\right)^{\gamma} \left(\frac{N}{n_j}\right)^{\frac{\gamma}{\theta}} n_j + \bar{w}\left(\frac{\bar{w}}{\delta W}\right)^{\gamma} N^{\frac{\gamma}{\theta}} n_j^{\frac{\theta-\gamma}{\theta}} \\
&= \left[(w_j^p)^{1+\gamma} + \frac{\bar{w}^{1+\gamma}}{\delta^{\gamma}}\right]\left(\frac{1}{W}\right)^{\gamma} N^{\frac{\gamma}{\theta}} n_j^{\frac{\theta-\gamma}{\theta}}
\end{aligned}
\tag{24}
$$

Using (18), we have that;

$$
\begin{aligned}
n_j &= \left(\frac{w_j}{W}\right)^{\theta} N \\
n_j^{\frac{1}{\theta}} &= \left(\frac{w_j}{W}\right) N^{\frac{1}{\theta}} \\
n_j^{\frac{1}{\gamma}} &= \left(\frac{w_j}{W}\right) N^{\frac{1}{\theta}} n_j^{\frac{-1}{\theta}} n_j^{\frac{1}{\gamma}} \\
w_j^{-1} n_j^{\frac{1}{\gamma}} &= \left(\frac{1}{W}\right) N^{\frac{1}{\theta}} n_j^{\frac{\theta-\gamma}{\theta\gamma}} \\
w_j^{-\gamma} n_j &= \left(\frac{1}{W}\right)^{\gamma} N^{\frac{\gamma}{\theta}} n_j^{\frac{\theta-\gamma}{\theta}}
\end{aligned}
\tag{25}
$$

Substituting (25) into (24),

$$
\begin{aligned}
w_j n_j &= \left[(w_j^p)^{1+\gamma} + \frac{\bar{w}^{1+\gamma}}{\delta^\gamma} \right] w_j^{-\gamma} n_j \\
w_j &= \left[(w_j^p)^{1+\gamma} + \frac{\bar{w}^{1+\gamma}}{\delta^\gamma} \right]^{\frac{1}{1+\gamma}}
\end{aligned}
\tag{26}
$$

Finally, using (18),

$$
\begin{aligned}
WN &= \int_0^1 w_j n_j dj \\
WN &= \int_0^1 w_j \left(\frac{w_j}{W} \right)^\theta N dj \\
W^{1+\theta} &= \int_0^1 w_j^{1+\theta} dj \\
W &= \left(\int_0^1 w_j^{1+\theta} dj \right)^{\frac{1}{1+\theta}}
\end{aligned}
\tag{27}
$$

Substituting $\bar{w}_j = \phi w_j^p$ into the definition of w_j,

$$
\begin{aligned}
w_j &= \left((w_j^p)^{1+\gamma} + \frac{(\bar{w}_j)^{1+\gamma}}{\delta^\gamma} \right)^{\frac{1}{1+\gamma}} \\
w_j &= \left((w_j^p)^{1+\gamma} + \frac{(\phi w_j^p)^{1+\gamma}}{\delta^\gamma} \right)^{\frac{1}{1+\gamma}} \\
w_j &= \left(1 + \frac{\phi^{1+\gamma}}{\delta^\gamma} \right)^{\frac{1}{1+\gamma}} w_j^p
\end{aligned}
\tag{28}
$$

Substituting $\bar{w}_j = \phi w_j^p$ in (20) and using (19), we obtain that

$$
\begin{aligned}
n_j^w &= \left(\frac{\bar{w}_j}{\delta W} \right)^\gamma \left(\frac{N}{n_j} \right)^{\frac{\gamma}{\theta}} n_j \\
&= \left(\frac{\phi}{\delta} \right)^\gamma \left(\frac{w_j^p}{W} \right)^\gamma \left(\frac{N}{n_j} \right)^{\frac{\gamma}{\theta}} n_j \\
&= \left(\frac{\phi}{\delta} \right)^\gamma n_j^p
\end{aligned}
\tag{29}
$$

And plugging (29) into the definition of n_j yields,

$$
\begin{aligned}
n_j &= \left((n_j^p)^{\frac{\gamma+1}{\gamma}} + \delta (n_j^w)^{\frac{\gamma+1}{\gamma}} \right)^{\frac{\gamma}{\gamma+1}} \\
n_j &= \left((n_j^p)^{\frac{\gamma+1}{\gamma}} + \delta \left(\left(\frac{\phi}{\delta} \right)^{\gamma} n_j^p \right)^{\frac{\gamma+1}{\gamma}} \right)^{\frac{\gamma}{\gamma+1}} \\
n_j &= \left(\left(n_j^p \right)^{\frac{\gamma+1}{\gamma}} \left(1 + \delta \left(\frac{\phi}{\delta} \right)^{\gamma+1} \right) \right)^{\frac{\gamma}{\gamma+1}} \\
n_j &= n_j^p \left(1 + \left(\frac{\phi^{1+\gamma}}{\delta^{\gamma}} \right) \right)^{\frac{\gamma}{\gamma+1}} \qquad (30)
\end{aligned}
$$

Using the definition of N and substituting (29);

$$
\begin{aligned}
N &= \left(\int_0^1 n_j^{\frac{\theta+1}{\theta}} \right)^{\frac{\theta}{\theta+1}} \\
&= \left(1 + \left(\frac{\phi^{1+\gamma}}{\delta^{\gamma}} \right) \right)^{\frac{\gamma}{\gamma+1}} N^p \qquad (31)
\end{aligned}
$$

We conjecture and presently verify:

$$
W^p = \left(\int_0^1 \left(w_j^p \right)^{1+\theta} \right)^{\frac{1}{1+\theta}} \qquad (32)
$$

Given the conjecture (32),

$$
W = \left(1 + \frac{\phi^{1+\gamma}}{\delta^{\gamma}} \right)^{\frac{1}{1+\gamma}} W^p \qquad (33)
$$

Substituting (30), (31), (32), (33) into (15),

$$
\begin{aligned}
W N^{\frac{-1}{\theta}} n_j^{\frac{1}{\theta}} n_j^{\frac{-1}{\gamma}} n_j^{\frac{\gamma+1}{\gamma}} &= w_j n_j \\
W^p \left(\frac{n_j^p}{N^p} \right)^{\frac{1}{\theta}} n_j^p &= w_j^p n_j^p \\
n_j^p &= \left(\frac{w_j^p}{W^p} \right)^{\theta} N^p \qquad (34)
\end{aligned}
$$

Now using the definition of $W^p N^p$ and (34)

$$
\begin{aligned}
W^p N^p &= \int_0^1 w_j^p n_j^p \\
&= \int_0^1 w_j^p \left(\frac{w_j^p}{W^p}\right)^{\theta} N^p \\
W^p &= \left(\int_0^1 \left(w_j^p\right)^{1+\theta}\right)^{\frac{1}{1+\theta}}
\end{aligned} \tag{35}
$$

verifying the conjecture.

6.2. Wage markdowns

We derive the markup equation (3) by differentiating the labor supply equation (1):

$$
\begin{aligned}
\frac{\partial logw_{ij}}{\partial logn_{ij}} &= \frac{\partial}{\partial logn_{ij}} \left\{ \frac{1}{\eta}\left[\log n_{ij} - logn_j^p\right] + \frac{1}{\gamma}\left[logn_j^p - logn_j\right] + \frac{1}{\theta}\left[logn_j - logN\right] + logW \right\} \\
&= \frac{1}{\eta} + \frac{\partial logn_j^p}{\partial logn_{ij}}\left(\frac{1}{\gamma} - \frac{1}{\eta}\right) + \frac{\partial logn_j}{\partial logn_{ij}}\left(\frac{1}{\theta} - \frac{1}{\gamma}\right)
\end{aligned}
$$

We now obtain the derivatives involved in deriving the markup equation (3). We use the definition of n_j^p and (22);

$$
\begin{aligned}
\frac{\partial logn_j^p}{\partial logn_{ij}} &= \frac{\partial n_j^p}{\partial n_{ij}} \frac{n_{ij}}{n_j^p} \\
&= \frac{n_{ij}}{n_j^p}\left[(n_j^p)^{\frac{-1}{\eta}} n_{ij}^{\frac{1}{\eta}}\right] \\
&= \frac{n_{ij} n_{ij}^{\frac{1}{\eta}}}{n_j^p (n_j^p)^{\frac{1}{\eta}}} \\
&= \frac{n_{ij} n_{ij}^{\frac{1}{\eta}}}{(n_j^p)^{\frac{1+\eta}{\eta}}} \\
&= \frac{n_{ij} n_{ij}^{\frac{1}{\eta}}}{\sum_i^{m_j} n_{ij}^{\frac{1+\eta}{\eta}}} \\
&= \frac{n_{ij}\left(\frac{w_{ij}}{w_j^p}\right)(n_j^p)^{\frac{1}{\eta}}}{\sum_i^{m_j} n_{ij}\left(\frac{w_{ij}}{w_j^p}(n_j^p)^{\frac{1}{\eta}}\right)} \\
&= \frac{n_{ij} w_{ij}}{\sum_i^{m_j} n_{ij} w_{ij}} \\
&= s_{ij}
\end{aligned} \tag{36}
$$

Using (34) and (36),

$$\begin{aligned}
\frac{\partial \log n_j}{\partial \log n_{ij}} &= \frac{\partial \log n_j}{\partial \log n_j^p} \frac{\partial \log n_j^p}{\partial \log n_{ij}} \\
&= \left(\frac{n_j^p}{n_j}\right) n_j^{-1/\gamma} \left(n_j^p\right)^{1/\gamma} s_{ij} \\
&= \left(\frac{n_j^p}{n_j}\right)^{1+\frac{1}{\gamma}} s_{ij} \\
&= \left\{ \left[1 + \left(\frac{\phi}{\delta}\right)^{\gamma+1} \right]^{-\frac{\gamma}{\gamma+1}} \right\}^{\frac{\gamma+1}{\gamma}} s_{ij} \\
&= \left(1 + \left(\frac{\phi^{1+\gamma}}{\delta^{\gamma}}\right)\right)^{-1} s_{ij} \qquad (37)
\end{aligned}$$

6.3. Aggregation

In market j, Berger et al. (2022) have the following six conditions (four firm-level optimality and two market-level aggregation):

$$y_{ij} = z_{ij} n_{ij}^{\alpha} \quad n_{ij} = \left(\frac{w_{ij}}{w_j}\right)^{\eta} n_j \quad w_{ij} = \mu_{ij} mrpl_{ij}$$

$$mrpl_{ij} = \alpha z_{ij} n_{ij}^{\alpha-1} \quad y_j = \sum_{i=1}^{m_j} y_{ij} \quad w_j = \left(\sum_{i=1}^{m_j} w_{ij}^{1+\eta}\right)^{\frac{1}{1+\eta}}$$

which deliver

$$y_j = \omega_j z_j n_j^{\alpha} \quad w_j = \mu_j \alpha z_j n_j^{\alpha-1}$$

where z_j, ω_j, μ_j have the appropriate expressions.

Across markets, Berger et al. (2022) have the following six conditions (four market-level aggregation and two economy-level aggregation)

$$y_j = \omega_j z_j n_j^{\alpha} \quad n_j^p = \left(\frac{w_j}{W}\right)^{\theta} N \quad w_j = \mu_j mrpl_j$$

$$mrpl_j = \alpha z_j n_j^{\alpha-1} \quad Y = \int_0^1 y_j dj \quad W = \left(\int_0^1 w_j^{1+\theta}\right)^{\frac{1}{1+\theta}}$$

which deliver

$$Y = \Omega Z N^{\alpha} \quad W = \mu \alpha Z N^{\alpha-1}$$

where Z, Ω, μ have the appropriate expressions.

Under the assumption $\bar{w}_j = \phi w_j^p$, we can obtain CES-type labor supply equations in the market economy. This makes the market economy in out model isomorphic to the economy set out in Berger et al. (2022). We exploit this property of the model to state equivalent equations for this model, and then use Berger et al. (2022)'s aggregation result.

In market j, we have the following six conditions (four firm-level optimality and two market-level aggregation):

$$y_{ij} = z_{ij} n_{ij}^{\alpha} \quad n_{ij} = \left(\frac{w_{ij}}{w_j^p}\right)^{\eta} n_j^p \quad w_{ij} = \mu_{ij} mrpl_{ij}$$

$$mrpl_{ij} = \alpha z_{ij} n_{ij}^{\alpha-1} \quad y_j = \sum_{i=1}^{m_j} y_{ij} \quad w_j^p = \left(\sum_{i=1}^{m_j} w_{ij}^{1+\eta}\right)^{\frac{1}{1+\eta}}$$

which deliver

$$y_j = \omega_j z_j \left(n_j^p\right)^{\alpha} \quad w_j^p = \mu_j \alpha z_j \left(n_j^p\right)^{\alpha-1}$$

where z_j, ω_j, μ_j have the appropriate expressions.

Across markets, we have the following six conditions (four market-level aggregation and two economy-level aggregation)

$$y_j = \omega_j z_j \left(n_j^p\right)^{\alpha} \quad n_j^p = \left(\frac{w_j^p}{W^p}\right)^{\theta} N^p \quad w_j^p = \mu_j mrpl_j$$

$$mrpl_j = \alpha z_j \left(n_j^p\right)^{\alpha-1} \quad Y = \int_0^1 y_j dj \quad W^p = \left(\int_0^1 \left(w_j^p\right)^{1+\theta}\right)^{\frac{1}{1+\theta}}$$

which deliver

$$Y = \Omega Z \left(N^p\right)^{\alpha} \quad W = \mu \alpha Z \left(N^p\right)^{\alpha-1}$$

where Z, Ω, μ have the appropriate expressions.

6.4. Model with homogenous firms

Assuming away productivity heterogeniety enables us to solve the model in closed form. Suppose each village consists of a single firm with productivity z and workfare. Using proposition 3, equilibrium firm-level output y, employment n, wage w, markdown μ are a solution to the following system of equations:

$$y = zn^{\alpha} \qquad w = \frac{\mu y}{n} \qquad \Phi^{1-\gamma\nu} w = n^{\nu} y^{\sigma} \qquad \mu = \frac{1}{1+\varepsilon^{-1}} \qquad \varepsilon^{-1} = \left(\frac{1}{\gamma} + \Phi^{-(1+\gamma)}\left(\frac{1}{\theta} - \frac{1}{\gamma}\right)\right)$$

where $\Phi = \left(1 + \frac{\phi^{\gamma+1}}{\delta^{\gamma}}\right)^{\frac{1}{\gamma+1}}$. Solving this system for n and w,

$$w = \Phi^{\gamma\nu-1} z^{\sigma} n^{\alpha\sigma+\nu} \quad n = \left[\Phi^{1-\gamma\nu} \mu z^{1-\sigma}\right]^{\frac{1}{\alpha(\sigma-1)+\nu+1}}$$

Taking logs,

$$logw = (\gamma\nu - 1)log\Phi + \sigma logz + (\alpha\sigma + \nu)logn$$
$$logn = \frac{1}{\alpha(\sigma - 1) + \nu + 1}\left[(1 - \gamma\nu)log\Phi + log\mu + (1 - \sigma)logz\right]$$

Differentiating with respect to $log\Phi$,

$$\frac{dlogw}{dlog\Phi} = (\gamma\nu - 1) + (\alpha\sigma + \nu)\frac{dlogn}{dlog\Phi}$$
$$\frac{dlogn}{dlog\Phi} = \frac{1}{\alpha(\sigma - 1) + \nu + 1}\left[(1 - \gamma\nu) + \frac{dlog\mu}{dlog\Phi}\right]$$

Since $\alpha(\sigma - 1) + \nu + 1 > 0$

$$\frac{dlogw}{dlog\Phi} > 0 \iff (\gamma\nu - 1)(1 - \alpha) + (\alpha\sigma + \nu)\frac{dlog\mu}{dlog\Phi} > 0$$
$$\frac{dlogn}{dlog\Phi} > 0 \iff (1 - \gamma\nu) + \frac{dlog\mu}{dlog\Phi} > 0$$

Restating,

$$\frac{dlogw}{dlog\Phi} > 0 \iff \gamma\nu > 1 - \frac{\alpha\sigma + \nu}{1 - \alpha}\frac{dlog\mu}{dlog\Phi}$$
$$\frac{dlogn}{dlog\Phi} > 0 \iff \gamma\nu < 1 + \frac{dlog\mu}{dlog\Phi}$$

Remains to calculate $\frac{dlog\mu}{dlog\Phi}$. Let $A = \Phi^{-(1+\gamma)}$ and $B = \frac{1}{\theta} - \frac{1}{\gamma}$. Then $\varepsilon^{-1} = \frac{1}{\gamma} + AB$. Differentiating,

$$\frac{dA}{d\Phi} = -(1 + \gamma)\frac{A}{\Phi} \implies \frac{d\varepsilon^{-1}}{d\Phi} = -(1 + \gamma)\frac{AB}{\Phi}$$

and hence

$$\frac{d\varepsilon^{-1}}{dlog\Phi} = -(1 + \gamma)AB$$

$$\frac{dlog\mu}{dlog\Phi} = \frac{-1}{1 + \varepsilon^{-1}}\frac{d\varepsilon^{-1}}{dlog\Phi}$$

$$\frac{dlog\mu}{dlog\Phi} = \frac{(1 + \gamma)AB}{1 + \varepsilon^{-1}}$$

where $A = \Phi^{-(1+\gamma)}$ and $\varepsilon^{-1} = \frac{1}{\gamma} + AB$.

In the competitive case, $\mu = 1$, so $\frac{dlog\mu}{dlog\Phi} = 0$. Consequently,

$$\begin{aligned}
\frac{dlogn}{dlog\Phi} &= \frac{1-\gamma\nu}{\alpha(\sigma-1)+\nu+1} \\
\frac{dlogw}{dlog\Phi} &= (\gamma\nu-1) + \frac{(\alpha\sigma+\nu)(1-\gamma\nu)}{\alpha(\sigma-1)+\nu+1} \\
&= \frac{(\gamma\nu-1)\alpha(\sigma-1)+(\gamma\nu-1)(\nu+1)+(\alpha\sigma+\nu)(1-\gamma\nu)}{\alpha(\sigma-1)+\nu+1} \\
&= \frac{(\gamma\nu-1)\alpha\sigma-(\gamma\nu-1)\alpha+(\gamma\nu-1)(\nu+1)-(\gamma\nu-1)\alpha\sigma-(\gamma\nu-1)\nu}{\alpha(\sigma-1)+\nu+1} \\
&= \frac{-(\gamma\nu-1)\alpha+(\gamma\nu-1)\nu+(\gamma\nu-1)-(\gamma\nu-1)\nu}{\alpha(\sigma-1)+\nu+1} \\
&= \frac{(\gamma\nu-1)(1-\alpha)}{\alpha(\sigma-1)+\nu+1}
\end{aligned}$$

6.5. Computation

1. Fix $\bar{w}_j$.
2. Enter the iteration with a guess of wages w_{ij}
3. Compute the private wage index:

$$w_j^p = \left(\sum_i w_{ij}^{\eta+1}\right)^{\frac{1}{\eta+1}}$$

4. Compute the village wage index:

$$w_j = \left(w_{pj}^{\gamma+1} + \delta^{-\gamma}\,\bar{w}_j^{\gamma+1}\right)^{\frac{1}{\gamma+1}}$$

5. Compute the aggregate wage index:

$$W = \left(\int_0^1 w_j^{1+\theta} dj\right)^{\frac{1}{1+\theta}}$$

6. Given the wedge $\phi_j = \frac{\bar{w}_j}{w_j^p}$, solve the markdown system to obtain $\{\mu_{ij}\}$.
7. Compute firm labor demand:

$$n_{ij} = \left(\frac{w_{ij}}{\alpha z_{ij}\mu_{ij}}\right)^{\frac{1}{\alpha-1}}$$

8. Aggregate to village output:

$$y_j = \sum_i z_{ij} n_{ij}^{\alpha}$$

9. Aggregate output:

$$Y = \int_0^1 y_j dj$$

10. Compute total labor supply:

$$N = \left(\frac{W}{Y^{\sigma}}\right)^{\frac{1}{\tau}}$$

11. Allocate labor across villages:

$$n_j = \left(\frac{w_j}{W}\right)^{\theta} N$$

12. Allocate private labor within village:

$$n_j^p = \left(\frac{w_j^p}{W}\right)^{\gamma} \left(\frac{N}{n_j}\right)^{\gamma/\theta} n_j.$$

13. Impose the within-private CES condition:

$$w_{ij,implied} = \left(\frac{n_{ij}}{n_j^p}\right)^{1/\eta} w_j^p,$$

14. Update $w_{ij,new} = w_{ij} + step * (w_{ij,implied} - w_{ij})$

15. Stop if $||w_{ij,new} - w_{ij}|| < tol$. Else set $w_{ij} = w_{ij,new}$.

6.6. Guner et al calibration

Table 2. Calibration Results (Guner et al., 2025)

	Interpretation / Moment	Model	Data
Targeted Moments			
workfare/market wage ratio		1	1
workfare/market labor ratio		0.57	0.57
mean HHI		0.099	0.03
labor share		0.52	0.52
markdown μ		0.53	0.54
workfare labor treatment effect		0.30	0.30
Parameters			
σ	risk aversion	1	
ξ	Pareto shape	6	
ν	inverse Frisch	0.1	
η	substitution across firms	10	
γ	substitution workfare vs private	10	
m	number of firms	11	
ϕ	workfare/private wage ratio	1	
δ	workfare disutility	1.05	
α	DRS	0.9	
θ	substitution across villages	0.08	
$\Delta\delta$	workfare improvement	-0.05	
Non-Targeted Moments			
market employment effect		0.025	0.18
market wage effect		0.028	0.10
employment effect with $\uparrow$ HHI		0.016	0.59

6.7. Brooks et al lower bound calibration

Table 3. Calibration Results (Kaboski et al., 2025 – Lower Bound)

	Interpretation / Moment	Model	Data
Targeted Moments			
workfare/market wage ratio		1	1
workfare/market labor ratio		0.57	0.57
mean HHI		0.26	0.03
labor share		0.26	0.52
markdown μ		0.31	0.30
workfare labor treatment effect		0.30	0.30
Parameters			
σ	risk aversion	1	
ξ	Pareto shape	6	
ν	inverse Frisch	0.1	
η	substitution across firms	9	
γ	substitution workfare vs private	10	
m	number of firms	4	
ϕ	workfare/private wage ratio	1	
δ	workfare disutility	1.05	
α	DRS	0.8	
θ	substitution across villages	0.08	
$\Delta\delta$	workfare improvement	-0.063	
Non-Targeted Moments			
market employment effect		0.043	0.18
market wage effect		0.039	0.10
employment effect with $\uparrow$ HHI		0.024	0.59

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